

2520090016

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Practical-6

The screenshot shows a Visual Studio Code editor window with a project named 'BookFlow'. The Explorer sidebar on the left shows the file structure: 'main.py' (selected), '_init_.py', 'catalog.py', and 'admin.py' under the 'BookFlow' directory. The main editor displays the code in 'main.py':

```
1 from fastapi import FastAPI
2 from routes.catalog import router as catalog_router
3 from routes.admin import router as admin_router
4
5 app = FastAPI(
6     title="BookFlow Library API"
7 )
8
9 app.include_router(catalog_router)
10 app.include_router(admin_router)
```

The bottom panel shows the Terminal with the following output:

```
uvicorn) (0.16.0)
Requirement already satisfied: colorama in C:\Users\palav\AppData\Roaming\Python\Python314\site-packages (from
PS C:\Users\palav\OneDrive\Desktop\DBMS P-6\BookFlow> pip install fastapi uvicorn
Requirement already satisfied: annotated-types>=0.6.0 in C:\Users\palav\AppData\Roaming\Python\Python314\site-p
ackages (from pydantic>=2.9.0->fastapi) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.5 in C:\Users\palav\AppData\Roaming\Python\Python314\site-pa
ckages (from pydantic>=2.9.0->fastapi) (2.46.5)
```

Swagger:

BookFlow Library API 0.1.0 OAS 3.1

/openapi.json

default

GET	/books	Get Books
GET	/books/{book_id}	Get Book
POST	/admin/books	Add Book
PUT	/admin/books/{book_id}	Update Book
DELETE	/admin/books/{book_id}	Delete Book

Schemas

Book > Expand all object

GET/books:

GET

/books

Get Books

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

curl -X 'GET' \ 'http://127.0.0.1:8000/books' \ -H 'accept: */*'

Request URL

http://127.0.0.1:8000/books

Server response

CodeDetails

200

Response body

{
 {
 "id": 101,
 "title": "Python Programming",
 "author": "John Smith",
 "price": 450,
 "isbn": "9781234567890",
 "status": "Available"
 },
 {
 "id": 102,
 "title": "Database Management",
 "author": "Robert Brown",
 "price": 550,
 "isbn": "9789876543210",
 "status": "Available"
 },
 {
 "id": 103,
 "title": "Advanced Operating Systems",
 "author": "Andrew Tanenbaum",
 "price": 750,
 "isbn": "9782222222222",
 "status": "Available"
 }
}

Download

GET/books/{book_id}:

GET

/books/{book_id}

Get Book

Parameters

Cancel

Name

Description

book_id * required

integer

(path)

101

Execute

Clear

Responses

Curl

curl -X 'GET' \ 'http://127.0.0.1:8000/books/101' \ -H 'accept: */*'

Request URL

http://127.0.0.1:8000/books/101

Server response

CodeDetails

200

Response body

{
 {
 "id": 101,
 "title": "Python Programming",
 "author": "John Smith",
 "price": 450,
 "isbn": "9781234567890",
 "status": "Available"
 }
}

Download

Response headers

content-length: 117
content-type: application/json
date: Fri, 18 Sep 2026 16:10:26 GMT
server: uvicorn

POST:

POST

/admin/books

Add Book

⌵

Parameters

Cancel

Reset

No parameters

Request body

required

application/json

⌵

Edit Value

Schema

{

"title": "Operating Systems",

"author": "Andrew Tanenbaum",

"price": 600,

"isbn": "9781111111111",

"status": "Available"

}

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/admin/books' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Operating Systems",
    "author": "Andrew Tanenbaum",
    "price": 600,
    "isbn": "9781111111111",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books

Server response

Code

Details

200

Response body

{

"message": "Book added successfully",

"book": {

"id": 104,

"title": "Operating Systems",

"author": "Andrew Tanenbaum",

"price": 600,

"isbn": "9781111111111",

"status": "Available"

}

}

Response headers

content-length: 169

content-type: application/json

date: Fri, 18 Sep 2026 16:11:51 GMT

server: uvicorn

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json

⌵

Controls Accept header.

Edit Value

Schema

PUT:

PUT

/admin/books/{book_id} Update Book

Cancel

Reset

Parameters

Name	Description
book_id * required	
integer (path)	103

Request body required

application/json

Edit Value | Schema

```
{  "title": "Advanced Operating Systems",  "author": "Andrew Tanenbaum",  "price": 750,  "isbn": "9782222222222",  "status": "Available"}
```

Execute

Clear

Responses

Curl

```
curl -X 'PUT' \  'http://127.0.0.1:8000/admin/books/103' \  -H 'accept: */*' \  -H 'Content-Type: application/json' \  -d '{  "title": "Advanced Operating Systems",  "author": "Andrew Tanenbaum",  "price": 750,  "isbn": "9782222222222",  "status": "Available"}
```

Curl

```
curl -X 'PUT' \  'http://127.0.0.1:8000/admin/books/103' \  -H 'accept: */*' \  -H 'Content-Type: application/json' \  -d '{  "title": "Advanced Operating Systems",  "author": "Andrew Tanenbaum",  "price": 750,  "isbn": "9782222222222",  "status": "Available"}
```

Request URL

http://127.0.0.1:8000/admin/books/103

Server response

Code	Details
200	Response body <pre>{ "message": "Book updated successfully", "book": { "id": 103, "title": "Advanced Operating Systems", "author": "Andrew Tanenbaum", "price": 750, "isbn": "9782222222222", "status": "Available" }}</pre> Download
	Response headers <pre>content-length: 180 content-type: application/json date: Fri, 18 Sep 2026 16:13:36 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

DELETE:

DELETE

/admin/books/{book_id} Delete Book

Parameters

Cancel

Name	Description
book_id * required integer (path)	103

Execute

Clear

Responses

Curl

```
curl -X 'DELETE' \
'http://127.0.0.1:8000/admin/books/103' \
-H 'accept: */*'
```

Request URL

```
http://127.0.0.1:8000/admin/books/103
```

Server response

Code	Details
200	Response body <pre>{ "message": "Book deleted successfully", "book": { "id": 103, "title": "Advanced Operating Systems", "author": "Andrew Tanenbaum", "price": 9.99, "isbn": "9782222222222", "status": "Available" } }</pre> Download Response headers <pre>content-length: 180 content-type: application/json date: Fri, 18 Sep 2026 16:14:51 GMT server: gunicorn</pre>